Supporting Information

Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible β-Sheet-like Order

Doga Ozgulbas[1], Miaoqi Chu[2], Alex Lavens[1], Nicholas Marks[1], Ming Du[2], Xiaobing Zuo[2], Michael Irvin[1], Heng Ma[1], Soenke Seifert[2], Byeongdu Lee[2], Rafael Vescovi[1], Ian Foster[1], Victor Mateevitsi[3], Don Jensen Jr.[2], Chase Akins[4], Arvind Ramanathan[1,*], Gyorgy Babnigg[4,*], and Qingteng Zhang[2,*]

[1]Data Science and Learning Division, Argonne National Laboratory, Lemont, Illinois 60439, United States
[2]X-ray Science Division, Argonne National Laboratory, Lemont, Illinois 60439, United States
[3]Argonne Leadership Computing Facility, Argonne National Laboratory, Lemont, Illinois 60439, United States
[4]Biosciences Division, Argonne National Laboratory, Lemont, Illinois 60439, United States

*ramanathana@anl.gov; gbabnigg@anl.gov; qzhang234@anl.gov

Supporting Information Contents

1 Recombinant Synthesis and Purification of $(VPAVG)_{30}$ 2
1.1 Materials . . . 2
1.2 Gene construction . . . 3
1.3 Protein expression and purification . . . 3

2 Sample Environments and X-ray Measurement Setup 3
2.1 Temperature-dependent SAXS/WAXS at 12-ID-B . . . 4
2.2 SA-XPCS at 8-ID-I . . . 4

3 12-ID-B SAXS/WAXS Analysis 5
3.1 Background subtraction and SAXS/WAXS merging . . . 5
3.2 Low-Q power law and WAXS peak fits . . . 5
3.3 Low-temperature unassembled state . . . 5

4 8-ID-I Absolute Scattering Cross Section 6
4.1 Absolute scattering cross section . . . 6
4.2 Transmission and incident-flux calibration . . . 6

5 8-ID-I SA-XPCS Correlation Analysis 6
5.1 Correlation functions and the fitted model . . . 6
5.2 Global fitting and uncertainties . . . 7
5.3 Stretching exponents and the fast-mode scaling . . . 8

6 Radiation-Damage Mitigation and Controls **9**

6.1 Fresh sample positions 9

6.2 Flux-dependence control 9

7 Macroscopic and Mesoscale Structural Reversibility **9**

7.1 Macroscopic reversibility 9

7.2 The seven-cycle experiment 9

7.3 Cycle-to-cycle repeatability 10

7.4 Dynamics at the cycled states 10

7.5 Cold-state artifacts and the outlier cut 10

8 Temperature Dependence of Arrest **11**

8.1 Measurements and reduction 11

8.2 Absolute scale, background and contrast 11

8.3 Fitting 13

8.4 Results 13

9 Code, Data and AI-Assisted Development **14**

9.1 Code and reduced-data availability 14

9.2 AI-assisted development 14

Video S1 and Figures S1–S11

1 Recombinant Synthesis and Purification of $(VPAVG)_{30}$

1.1 Materials

Shrimp alkaline phosphatase, PfuUltra II Fusion HS DNA polymerase, GoTaq Green Master Mix, and T4 DNA ligase with PEG4000 additive were purchased from Thermo Fisher Scientific. Exonuclease I was purchased from New England Biolabs. CircLigase ssDNA ligase was purchased from Lucigen, and KOD Hot Start DNA polymerase was purchased from Sigma-Aldrich. PCR products and plasmids were purified using Monarch PCR and DNA Cleanup kits and Monarch Plasmid Miniprep columns (New England Biolabs). DNA was recovered from agarose gels using the Monarch DNA Gel Extraction kit (New England Biolabs). Oligonucleotides and the 5′-phosphorylated synthetic template used for rolling-circle amplification were obtained from Integrated DNA Technologies. *E. coli* BL21(DE3)-Gold cells were obtained from Stratagene.

1.2 Gene construction

The ELP gene was prepared by overlap-extension rolling-circle amplification (OERCA).[1] A single-stranded DNA template encoding five VPAVG repeats was circularized in a 20 µL reaction containing 10 pmol template, 2 µL of 10× CircLigase buffer, 1 µL of 1 mmol L^{-1} ATP, 1 µL of 50 mmol L^{-1} $MnCl_2$, and 100 U CircLigase. The reaction was incubated for 1 h at 60 °C, heat-inactivated for 10 min at 80 °C, purified, and treated with exonuclease I for 1 h at 37 °C to remove unreacted linear template.

OERCA was performed with 300 ng circular template, 1.5 µL each of 50 µmol L^{-1} primers A1 and A2 (Figure S1a), 2 µL of 70% GC dNTP mix, 2 µL of 25 mmol L^{-1} $MgSO_4$, 10 µL of 10× PfuUltra II buffer, 5 µL DMSO, and 2 µL PfuUltra II polymerase in a total volume of 100 µL. Cycling consisted of 2 min at 95 °C; 30 cycles of 20 s at 95 °C, 20 s at 54 °C, and 90 s at 72 °C; and a final 10 min at 72 °C. Products above approximately 1.2 kb were isolated from a 0.8% agarose gel.

Ligation-independent cloning adapters were introduced by PCR using primers B1 and B2 (Figure S1a), and the products were inserted into the pMCSG53 vector.[2, 3] Constructs were transformed into *E. coli* BL21(DE3)-Gold, screened for soluble expression, and sequence verified by Sanger sequencing. The selected construct encodes the amino-acid sequence `MHHHHHHSSGVDLGTENLYFQSNA(VPAVG)30`, comprising a cleavable N-terminal affinity/fusion sequence followed by 30 VPAVG repeats. That leader was kept rather than cleaved, because its single tyrosine is the only chromophore in the chain and the A_{280} concentration measurement depends on it. Every measurement reported here is therefore on the tagged construct, and the leader accounts for 24 of the 174 residues and holds all eight charged residues.

1.3 Protein expression and purification

Fresh transformants were grown in Terrific Broth containing 10 mmol L^{-1} $MgSO_4$ and 300 µg mL^{-1} ampicillin at 37 °C to an optical density of 0.6–0.8 at 600 nm. Cultures were cooled to 18 °C, induced with 0.2 mmol L^{-1} IPTG, and grown overnight. Cells were harvested at 5000 × g, resuspended in ice-cold lysis buffer (500 mmol L^{-1} NaCl, 5 vol% glycerol, 50 mmol L^{-1} 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES) at pH 8.0, 20 mmol L^{-1} imidazole, and 10 mmol L^{-1} 2-mercaptoethanol), treated with 1 mg mL^{-1} lysozyme for 10 min, sonicated, and clarified at 29,000 × g.[4]

The supernatant was applied to 5 mL IMAC Sepharose High Performance resin, washed with lysis buffer, and eluted with lysis buffer containing 500 mmol L^{-1} imidazole. Purification fractions were monitored by SDS-PAGE. The purified protein was exchanged into 20 mmol L^{-1} HEPES at pH 8.0, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} dithiothreitol (DTT) and concentrated using a 10 kDa molecular-weight-cutoff centrifugal concentrator. Protein concentration was determined by direct A_{280} measurement on a NanoDrop spectrophotometer using the sequence-specific molar extinction coefficient applied during sample preparation (calculated molecular weight 15,461 Da and ε_{280} = 1490 M^{-1} cm^{-1}, both for the tagged construct). The final concentration was approximately 100 mg mL^{-1}; samples were stored at 4 °C, and freshly prepared protein was used for the X-ray measurements. The expression, solubility, and purity of the protein were evaluated by SDS-PAGE (Figure S1b).

2 Sample Environments and X-ray Measurement Setup

2.1 Temperature-dependent SAXS/WAXS at 12-ID-B

Temperature-dependent SAXS/WAXS was performed at Advanced Photon Source (APS) beamline 12-ID-B. Two aliquots of $(VPAVG)_{30}$ at approximately 100 mg mL^{-1} in 20 mmol L^{-1} HEPES (pH 8), 250 mmol L^{-1} NaCl and 2 mmol L^{-1} DTT, nominally the same concentration used for the 8-ID-I measurements, were sealed in 2 mm quartz capillaries. The two beamtimes fell in different APS run cycles several months apart, and because the protein is purified fresh for each experiment, the 12-ID-B and 8-ID-I material came from separate preparations of the same construct. The aliquot designated Measurement underwent more than 10 heating/cooling cycles across the thermal transition, whereas the Reference aliquot remained refrigerated at 4 °C. Both aliquots were subsequently measured at 10 °C and 30 °C during APS run 2021-1. The 13.3 keV beam had an approximate size of 100×150 μm^2 and an incident flux of approximately 5×10^{11} photons s^{-1}. SAXS was recorded with an Eiger2 9M detector at an approximately 2 m sample-to-detector distance, the same energy and camera length used for a published SAXS measurement at this beamline[5] and the same camera length as another[6], while WAXS was recorded with a Pilatus 300K detector at 0.42 m. The Pilatus detector was mounted above the direct-beam path, allowing the transmitted beam to continue to the SAXS detector so that SAXS and WAXS were acquired simultaneously with a 0.1 s exposure. Buffer and capillary backgrounds were subtracted, and the SAXS and WAXS profiles were stitched over their common Q interval. At each set point the capillary was held at the target temperature for at least 10 min before the exposures were started, so each profile reports the structure present after at least that long at temperature rather than during the temperature change. The two experiments used different sample cells, thermal paths, and thermometry, and the time spent at 30 °C before each 12-ID-B exposure is known only as a lower bound. The 12-ID-B 30 °C profiles are therefore kept off the isothermal waiting-time axis of the 8-ID-I experiment; they establish the low- and high-temperature structures and their reversibility, not the kinetics of assembly.

2.2 SA-XPCS at 8-ID-I

Small-angle X-ray photon correlation spectroscopy (SA-XPCS) was performed at APS 8-ID-I. Approximately 100 mg mL^{-1} $(VPAVG)_{30}$ in the same buffer was loaded into an aluminum cell containing a 3 mm-diameter, 2 mm-thick liquid chamber sealed by 0.127 mm polycarbonate windows (Figure S2). The cell was inserted into an anodized-aluminum holder in one temperature zone of the Quantum Northwest 3×3 multi-sample temperature control unit, the same stage used in Ref. 7. Each zone is driven by a Peltier element and a resistive heater, and a calibrated resistance-temperature detector (RTD) embedded adjacent to the Peltier stage supplies the controller feedback and readout. The aluminum holder and aluminum cell provided direct conductive thermal coupling. A partially coherent 10.94 keV X-ray beam with 0.03% FWHM bandwidth and incident flux 1.2×10^{10} photons s^{-1} was focused to approximately 10×10 μm^2. Coherent scattering was recorded 7.8 m downstream with a Rigaku XSPA-500k photon-counting detector comprising 1024×512 pixels with a 76 µm pixel size and a 320 µm-thick silicon sensor. The detector was operated in 2-bit mode at a 50 kHz frame rate; each acquisition contained 100,000 frames, corresponding to 20 µs per frame and 2 s total duration. The multi-tau delay list is built from the recorded 20 µs frame time, and the longest accessible delay is 1.638 s.[8]

Every 8-ID-I measurement reported here follows the same acquisition pattern. Single 2 s acquisitions were taken one after another, each at a previously unexposed position on a grid of sample positions, at a cadence of roughly 6 s per acquisition once stage motion and readout are included. A single acquisition is therefore a snapshot of a fresh piece of sample at a known time, and a series steps through increasing time at that cadence. SAXS profiles were obtained by temporal and azimuthal averaging. Every reported

quantity is a group average rather than a single acquisition: consecutive acquisitions were averaged in contiguous run ranges after the two outlier cuts of Section 7.5, and the quoted time is the acquisition time of the first run in the range. The two thermal protocols are given with their results, the seven-cycle experiment in Section 7.2 and the isothermal temperature series in Section 8.1.

3 12-ID-B SAXS/WAXS Analysis

3.1 Background subtraction and SAXS/WAXS merging

The beamline-reduced `.avg` files contain Q, $I(Q)$ and a pointwise intensity uncertainty, the counting error $\sqrt{I}$. For each aliquot and temperature the capillary-plus-buffer profile was subtracted with a scale factor α of 1.00 for SAXS and the empirical 0.95 used in the original WAXS reduction, and the uncertainties were propagated through that subtraction as $\sigma = \sqrt{\sigma_{\mathrm{s}}^2 + \alpha^2 \sigma_{\mathrm{b}}^2}$. The WAXS intensity and its uncertainty were then multiplied by the ratio of the mean SAXS and WAXS intensities over their common Q interval, that ratio being treated as exact, and the two profiles were concatenated and sorted by Q.

Those propagated uncertainties enter one result. The Guinier fit of Section 3.3 is weighted, each point carrying $1/\sigma_{\ln I}^2$ with $\sigma_{\ln I} = \sigma_I / I$, and they therefore set the quoted errors on R_g and I_0. The low-Q power-law fit and the two Gaussian peak fits are unweighted, so the exponent and the peak centers and widths do not depend on them. Neither the 0.95 nor the overlap ratio affects a reported result either: no absolute WAXS intensity is used anywhere in this work, and the centers and widths of the two peaks, and hence the β-sheet-like spacings, depend on neither.

3.2 Low-Q power law and WAXS peak fits

For Figure 2 of the main text, the low-Q slope was obtained by an unweighted linear fit of $\log_{10} I$ against $\log_{10} Q$ over $0.012 < Q < 0.040$ Å^{-1} for the Reference aliquot at 30 °C. The two WAXS features were fitted independently to a Gaussian plus linear background over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å^{-1}, respectively. The low-Q exponent is reported as an exponent and is deliberately not converted into a fractal dimension. Two conditions set out by Teixeira[9] are not met. First, a mass-fractal dimension must satisfy $D < 3$, whereas the fitted 12-ID-B slopes are -3.17 and -3.15; exponents between 3 and 4 belong instead to the surface-fractal form $Q^{-(6-D_{\mathrm{s}})}$, which that reference notes is not easily distinguished from the Q^{-4} of a smooth interface. Second, a fractal description is expected to hold over at least one decade in Q, and the fit window $0.012 < Q < 0.040$ Å^{-1} spans 0.52 decades. The reduced profiles and scripts implementing the background subtraction, overlap scaling, Guinier analysis, power-law fit, and peak fits are available in the public repository identified in Section 9.

3.3 Low-temperature unassembled state

The Reference profile at 10 °C contains a Guinier regime satisfying $Q_{\max} R_g = 1.246$ (Figure S3). A fit of $\ln I(Q)$ against Q^2, weighted by those propagated uncertainties, gives

$$R_g = 22.65 \pm 0.50 \text{ Å}. \tag{S1}$$

For reference, the cross-β pattern is reported at approximately 4.7–4.8 Å along the fibril direction and between 8 and 11 Å perpendicular to it.[10, 11] Our 8.8 Å falls inside that second range and our 4.5 Å sits just below the first.

The fitted size is molecular in scale and is consistent with a dispersed, unassembled low-temperature state rather than persistent mesoscale aggregates. Because the protein concentration is high, this value is reported as an apparent R_g; weak intermolecular structure-factor contributions cannot be excluded. Recovery of the same low-temperature profile after repeated cycling establishes that the large high-temperature assembly is no longer detectable after cooling.

4 8-ID-I Absolute Scattering Cross Section

4.1 Absolute scattering cross section

For the 8-ID-I data, the differential scattering cross section was calculated as

$$I(Q) = A_s I_s(Q) - A_b I_b(Q), \tag{S2}$$

where $I_s(Q)$ and $I_b(Q)$ are the temporal-mean detector intensities from the sample and from its background, per pixel and per frame. The background is a buffer where one was measured and the empty cell of Section 8.2 otherwise. The conversion factors were

$$A_j = \frac{N_\mathrm{f}}{F_j T_j t_\mathrm{exp} L \Delta\Omega}, \qquad j \in \{s, b\}, \tag{S3}$$

where F_j is the incident photon flux, T_j is the transmission, $N_\mathrm{f} = 100{,}000$ is the number of frames, $t_\mathrm{exp} = 2\,\mathrm{s}$ is the total exposure time, so that $t_\mathrm{exp}/N_\mathrm{f}$ is the 20 µs frame time, $L = 2\,\mathrm{mm}$ is the sample thickness, and $\Delta\Omega = (d_\mathrm{pix}/D)^2$ is the solid angle per detector pixel.[12] Lengths are carried in millimeters throughout, so $A_j I_j$ is in mm^{-1}, and that is the quantity plotted in Figures 3a, S4b and S7–S10: no further scaling of any kind is applied between the calibration and the figure.

4.2 Transmission and incident-flux calibration

Sample transmission was obtained from ion chambers immediately upstream and downstream of the sample. Across the attenuation series, the transmission of the intervening air path was independent of beam intensity within uncertainty, with a mean value of 0.8679 ± 0.0009 (Figure S5a; mean ± standard deviation). The upstream ion-chamber signal was converted to incident photon flux by calibration against a PIN diode positioned in the flight path upstream of the beamstop. The measured photon flux varied linearly with the upstream ion-chamber readout (Figure S5b).

5 8-ID-I SA-XPCS Correlation Analysis

5.1 Correlation functions and the fitted model

The intensity autocorrelation function was calculated with a multi-tau algorithm,

$$g_2(Q, \Delta t) = \frac{\langle I(Q,t) I(Q, t+\Delta t)\rangle_t}{\langle I(Q,t)\rangle_t^2}. \tag{S4}$$

The implementation is the 8-ID-I reduction workflow; its numerical treatment of the multi-tau scheme at these frame rates is described in detail elsewhere.[8, 13]

The reduction produced 27 azimuthally averaged dynamic Q bins of nominal width 9.9×10^{-4} Å^{-1} spanning 0.003 59–0.029 37 Å^{-1}, and 61 delay times from 20 µs to 1.64 s. The five lowest bins, centered at $Q = 0.00359$, 0.00458, 0.00557, 0.00657 and 0.00756 Å^{-1} and corresponding to $2\pi/Q$ of approximately 175, 137, 113, 96 and 83 nm, are analyzed here. The restriction is a signal-to-noise limit: the scattered intensity falls by more than an order of magnitude between the lowest bin and 0.00756 Å^{-1}, so above that bin the correlation function is dominated by counting noise. Figure S4c shows the still-mobile limit of the two-step behavior directly.

The measured g_2 carry two decays separated by several decades in delay, and were parameterized through a normalized intermediate scattering function g_1 built from two stretched exponentials,

$$\begin{aligned} g_2(Q,\Delta t,t_w) &= 1 + \beta\,|g_1(Q,\Delta t,t_w)|^2\,, \\ g_1(Q,\Delta t,t_w) &= f(Q,t_w)\exp\!\left[-\left(D_0(t_w)\,Q^2\,\Delta t\right)^{p_{\mathrm{fast}}}\right] \\ &\quad + [1-f(Q,t_w)]\exp\!\left[-\left(\frac{\Delta t}{\tau_{\mathrm{slow}}(Q,t_w)}\right)^{p_{\mathrm{slow}}(t_w)}\right]. \end{aligned} \tag{S5}$$

The first line is the Siegert relation. The two-mode form on the second line is the same one used for thermoreversible nanoparticle gels, where the two-step decay is fitted as a linear sum of two stretched exponential modes with complementary amplitudes a and $1-a$.[14] The same two-mode decomposition, with a fast decay and a slow stretched decay whose amplitude sets the intermediate plateau, describes ergodicity breaking in aging nanoclay suspensions.[15] Here g_1 is the normalized intermediate scattering function, f is the fitted fraction of fast dynamics and $1-f$ its complement. β is the instrumental speckle contrast, set by the transverse coherence of the beam and by the ratio of speckle size to pixel size. It is a property of the beamline and the detector rather than of the sample, is measured on a static reference, and is held fixed at 0.141 throughout (Section 8.2). τ_{slow} is free at each Q and each t_w. The fast relaxation time is not free, being tied to the diffusive form $\tau_{\mathrm{fast}}(Q,t_w) = 1/[D_0(t_w)Q^2]$, so one coefficient replaces five free times at each waiting time. The Kohlrausch–Williams–Watts exponents carry no Q argument because they are shared across the five bins. p_{fast} is free at each waiting time rather than shared along a hold: these holds run from a mobile state into arrest, p_{fast} falls from about 1 to about 0.2 as they do, and a single shared value is rejected by an F test at $p < 10^{-16}$ in every loading that arrests. The decomposition is phenomenological: it describes the observed first decay and persistent plateau without assigning either mode to a unique molecular mechanism.

5.2 Global fitting and uncertainties

Equation S5 was fitted to all five Q bins of one averaged group at once. f and τ_{slow} are free in each bin; D_0, p_{fast} and p_{slow} are single values shared across the bins of that group. Groups are fitted independently of one another, so nothing is shared along a hold. Fits minimized the error-weighted residuals $[g_{2,\mathrm{model}} - g_{2,\mathrm{data}}]/\sigma_{g_2}$ with `scipy.optimize.least_squares`, using the pointwise g_2 uncertainties stored in the reduced files, so each delay enters with weight $1/\sigma_{g_2}^2$ and the noisy short delays count for little. Bounds were wide ($0 \le f \le 1$, $1 \le \tau_{\mathrm{slow}} \le 10^4$ s, $0.2 \le p \le 3$, $10^2 \le D_0 \le 10^{10}$ Å^2 s^{-1}). Each fit is started from the minimum of a p_{fast} profile so that it cannot settle in a solution where the fast term imitates the slow one. Reduced χ^2 runs from 0.90 to 3.34 over the four loadings, and a group is kept only if it stays below 10. Starting values and the remaining numerical settings are in the code (Section 9).

Parameter uncertainties are the square roots of the diagonal of the covariance matrix, estimated as the Moore–Penrose pseudo-inverse of $J^{\mathrm{T}}J$ with J the Jacobian of the error-weighted residuals at the solution; the pseudo-inverse is used so that a parameter refining to a bound does not leave the uncertainties of the

remaining, well-constrained parameters undefined. The fast fraction reaches its upper bound of one in 14 of the 170 bin fits, all of them in still-mobile groups of the 28.00 and 28.25 °C holds; its smallest fitted value anywhere is 0.07. These are the error bars plotted in Figure 3c and in Figures S7–S10.

The slow component remains only partially relaxed during the 2 s acquisition and frequently yields $\tau_{\mathrm{slow}} > 2$ s, so it is an effective extrapolation rather than a sampled terminal relaxation time. Conclusions about arrest therefore rest on directly observed features: the decreasing amplitude of the first decay, the rising plateau, and the systematic decrease in f. XPCS does not determine where the arrested regions are, $1 - f$ is not a geometric volume fraction because scattering contributions are weighted by structure and contrast, and because the measurement does not reach infinite delay it is not the formal nonergodicity factor either. Panels (d)–(g) of Figures S7–S10 give the correlation functions and fits at the four additional Q values: the same two-step shape and the same monotonic loss of the fast step appear at every measured Q, so the trend of Figure 3c belongs to the whole 83–175 nm range and not to a single bin.

5.3 Stretching exponents and the fast-mode scaling

Both stretching exponents stay below unity. p_{fast} runs from 0.55 to 1.00 while a hold is still mobile and from 0.20 to 0.51 once it has arrested, and p_{slow} is free as well. That the breadth of the fast relaxation grows as the material locks up is itself a measurement rather than a modelling choice: a single p_{fast} shared along a hold is rejected by an F test at $p < 10^{-140}$ in each of the three loadings that arrest, while in the 28.00 °C loading, which does not arrest, freeing it moves the reduced χ^2 only from 1.21 to 1.19. Stretched decays of this kind indicate a broad distribution of relaxation times rather than a single process, and are characteristic of protein and colloidal gels,[14, 16] although the direction of the departure from $p = 1$ varies between systems: the egg-white gel of Ref. 16 relaxes with compressed rather than stretched exponentials, the signature of stress-driven rather than diffusive relaxation.

The fast relaxation occurs on approximately 10^{-4} to 10^{-3} s and becomes faster with increasing Q. Equation S5 imposes the diffusive scaling $\tau_{\mathrm{fast}} \propto Q^{-2}$ rather than fitting it, and a stretched g_2 remains compatible with diffusion provided p does not depend on Q, which holds here by construction. Releasing the constraint, so that τ_{fast} is free in every bin, returns a median Q exponent of -2.5 over the 32 groups whose fast amplitude at the lowest Q exceeds 0.15, with 80% of them between -2.6 and -1.3. That release is also the general two-mode form of Ref. 14, in which the fast time is free in every bin rather than carried by one coefficient. Refitting every group that way moves f at the lowest Q by 0.006 at the median, by 0.005 or less in the groups where the fast mode has nearly gone, and by at most 0.16 in a single group, and it moves the scaling temperature of Figure 4 from 0.224 to 0.220 K, a shift of 1.7% against a bin-to-bin spread of 5%. Those fits are seeded as every other fit here is, from the minimum of a p_{fast} profile. That matters, because with p_{fast} free the two-mode form has local minima in which the fast term imitates the slow one: started instead from the bare default of 0.5, two of the 34 groups settle in one of them at reduced χ^2 of 28 and 93, against 2.6 and 9.2 seeded, and the shift turns the other way, to +0.9%. Both are small against the spread across Q. The collapse therefore does not rest on the diffusive constraint. Where it is not, that is in the later groups of the holds that arrest furthest, the exponent collapses toward zero because τ_{fast} stops being constrained once the fast decay has gone, not because the relaxation changes character. Those exponents bracket the imposed value of -2 but sit systematically steeper than it, so the diffusive form is reported here as an imposed approximation that the data are consistent with, and not as a fitted result. What is read from it is the fitted coefficient D_0 of Section 8.3, and the conclusions of this work rest on f rather than on D_0: f is set by the height of the

plateau and stays measured to the end of every hold, whereas any fast-mode time scale needs a resolved decay. Freezing p_{fast} at 0.5 for every group, against fitted values spanning 0.20 to 1.00, moves f at the lowest Q by at most 0.07 on a range of 0.07 to 0.99 and the scaling temperature of Figure 4 by 3%, so neither result depends on how the stretching is treated.

6 Radiation-Damage Mitigation and Controls

6.1 Fresh sample positions

The grid of positions referred to in Section 2.2 was Cartesian, 100×100 points in the aluminum cell, with adjacent points separated by at least one beam diameter. Because no position was visited twice, cumulative dose cannot track the elapsed time of the isothermal experiment, so the trend in the fitted fast fraction cannot be a dose artifact.

6.2 Flux-dependence control

In addition, to show that there is no significant radiation-induced artifact, an independent dose-rate test was performed after the APS Upgrade using a $20\,\mathrm{mg\,mL^{-1}}$ $(\mathrm{VPAVG})_{30}$ sample. Every acquisition was taken at a stage-RTD readout of 33.0 °C, above the transition temperature, so the control probed the assembled high-temperature state rather than the low-temperature dispersed one. Attenuation selected incident fluxes from 2.7×10^{11} to 4.9×10^{11} photons $\mathrm{s^{-1}}$, approximately 23–41 times the flux used for the main experiment. The SAXS line shapes and g_2 decays showed no systematic flux dependence (Figure S6). That control sample is five times more dilute than the $100\,\mathrm{mg\,mL^{-1}}$ material of the main experiment, and it was measured with the post-Upgrade beam. The main series is protected instead by the fresh position used for every acquisition (Section 6.1).

7 Macroscopic and Mesoscale Structural Reversibility

7.1 Macroscopic reversibility

Video S1 provides a direct visual demonstration of thermal reversibility. Brief warming of a capillary containing $(\mathrm{VPAVG})_{30}$ produces turbidity within approximately 5 s, indicating structures large enough to scatter visible light. After removal of the heat source, the sample returns to optical transparency in approximately 60 s. This recovery is consistent with the overlap of the cycled Measurement and refrigerated Reference SAXS/WAXS profiles at 10 °C. Optical transparency does not specify molecular structure, but independently confirms dissolution of the mesoscale assembly on cooling.

7.2 The seven-cycle experiment

The reversibility comparison of Figure 2 sets two separate 12-ID-B aliquots against each other. The test in Figure S4 is more direct: one aliquot at 8-ID-I, taken through seven consecutive cycles and compared with itself, so that any difference between cycles can only be irreversibility. The aliquot is a separate loading of the same stock, measured in the 2022-1 beamtime that also supplies the absolute-scale calibration of Section 4. Each cycle held the sample at 6 °C, ramped to 34 °C at $1.01\,^{\circ}\mathrm{C\,min^{-1}}$ over 270 consecutive

2 s acquisitions, and returned to 6 °C, taking 5.8 h in total. No acquisitions were taken during cooling, so the cooling segments of Figure S4a are a projection from the fixed control protocol rather than a measurement; every solid segment is measured, once per acquisition, from the RTD of the stage zone holding the cell.

7.3 Cycle-to-cycle repeatability

Three states are sampled in every cycle: 6 °C, and two windows near the top of each ramp at 31.9 °C and 33.8 °C. Averaged over $0.004 < Q < 0.008$ Å^{-1}, the seven cycles agree to a relative standard deviation of 9% at 6 °C and 5% at 33.8 °C, with no drift from the first cycle to the seventh. At 31.9 °C the spread is 33%, but that reflects where on the ramp each cycle happened to be sampled rather than a difference between cycles: the windows fall between 31.62 °C and 32.48 °C, which is on the steep part of the transition. Regressing the band-averaged $\ln I$ of each cycle on that cycle's mean window temperature gives $\mathrm{d}\ln I/\mathrm{d}T = 0.95\,^{\circ}\mathrm{C}^{-1}$ with $r = 0.974$, and the spread taken about that line falls to 7%. The same regression at 33.8 °C, past the steep part, takes 5% to 1%. The cycles therefore repeat to better than 10% at every state once the sampling temperature is accounted for.

7.4 Dynamics at the cycled states

The correlation functions say what state the sample was in when each was taken (Figure S4c). At 33.8 °C the correlation function shows no resolvable decay anywhere in the accessible window, from 10^{-2} to 2 s, in any of the seven cycles: the sample was arrested on every heating. At 31.9 °C it decays throughout that same window, so the assembly is still mobile 1.9 °C lower. The dashed line in that panel is the measured correlation level of a static reference recorded in the same beamtime, read at the same Q bin. A static sample cannot relax, so its level is the instrumental contrast, and showing it as a curve rather than as a fitted number keeps the comparison between two measurements. The 33.8 °C correlation functions sit just below it and the 31.9 °C ones fall well away from it. A contrast value is needed to fit a correlation function, which is why the holds of Section 8 carry their own calibration; it is not needed to judge whether a sample has arrested, which is what this panel does, so no arrested fraction is quoted from it. No correlation function is shown at 6 °C, where the coherent signal at the lowest Q is too weak for any contrast to be resolved.

7.5 Cold-state artifacts and the outlier cut

Two features of the 6 °C state need comment. First, its profile still rises toward low Q instead of reaching the flat Guinier plateau of a fully dispersed solution. Mis-subtracted solvent does not account for it: the buffer contributes only about 35% of the signal at the lowest Q, so rescaling it to remove the rise would subtract most of the low-Q signal and would degrade the cycle-to-cycle agreement rather than improve it. Second, the cold state is where transient scatterers appear. Taking every acquisition as recorded, the intensity at $Q = 0.0035$ Å^{-1} within a single 6 °C group is strongly right-skewed, while the same acquisitions at $Q = 0.03$ Å^{-1} are symmetric at ordinary counting statistics; a long positive tail confined to low Q is the signature of a large object crossing the 10×10 μm^2 beam during a minority of acquisitions. The cross-correlation cut does not catch them, because it compares curve shape over the whole Q range and a curve anomalous in three low-Q bins alone stays close to the group mean by that measure.

A second, robust cut removes them, and is applied to every group in the paper on identical terms

with no per-group list. For each acquisition the mean of $I(Q)$ over $0.004 < Q < 0.008$ Å^{-1} is compared with the group median, and the acquisition is rejected if it exceeds the median by more than three robust standard deviations, the scale being 1.4826× the median absolute deviation and both statistics iterated to convergence. The test is one-sided because a transient scatterer only adds intensity. Across the groups of this experiment it removes 33 of the 512 acquisitions surviving the cross-correlation cut, 6.4%: 4, 2, 1, 11, 4, 1 and 1 from the seven 6 °C groups, 2 and 7 from the two buffer groups, and none at all from any of the fourteen high-temperature windows, which is consistent with the skew being confined to the cold state. It shifts one absolute-scale coefficient by 0.03% and leaves every fitted correlation-function parameter unchanged. Its effect here is reported rather than absorbed: the cycle-to-cycle spread at 6 °C falls from 13% to 9%, and what remains is a mild rise over the last two cycles rather than one anomalous group. The threshold is not tuned, the same acquisitions being selected for any value between 2.5 and 3 and for two other choices of band.

8 Temperature Dependence of Arrest

8.1 Measurements and reduction

The four loadings that carry the main text were measured at 8-ID-I in October 2021, with the beam energy, detector, detector distance and cell design of Section 2.2. They are the same construct as the 12-ID-B material, from a separate preparation. After a reference measurement at 6 °C, each loading was heated from 20 °C at 0.1 °C min^{-1} to its final temperature and held there: loading D8 at 28.00 °C, D6 at 28.25 °C, D7 at 28.50 °C and D9 at 28.75 °C. The temperatures are the readings of the stage RTD, and $t_w = 0$ is the first acquisition with the RTD at its target. Acquisitions of 100,000 frames ran at fresh positions through the ramp and then for 7100, 8500, 3900 and 2200 s of the hold, respectively.

The 2021 Data Exchange records were converted to the NeXus format with the incident- and transmitted-beam monitors and the ring current carried across unchanged, and the frames were correlated with the 8-ID-I reduction workflow of Section 5. Consecutive acquisitions were averaged in groups of 100, about 10 min each, with one group boundary at $t_w = 0$, through the same two outlier cuts that every other group in this work receives (Section 7.5). One detector pixel read a constant 1.5 photons per frame in the measurements of 17 October and is masked in the Q map the series is correlated against. It lay at $Q = 0.0214$ Å^{-1}, outside the five Q bins of the correlation analysis; masking it left every fitted quantity unchanged.

8.2 Absolute scale, background and contrast

The beamtime used the same ion chambers as the calibration of Section 4, so $I(Q)$ is put on that absolute scale, each group with its own monitor readings. The four loadings transmit 0.383 to 0.389. The background is a separate measurement of the sample cell, subtracted on the absolute scale exactly as the buffer is in Figure S4b. Six candidate measurements are available, four of them recorded on the same day as the loadings. They are identified here by what they transmit and what they scatter rather than by the labels entered at the beamline, which do not reliably distinguish a cell left empty from one holding buffer. Four of the six recorded a transmitted-beam monitor. One of those passes 0.94 of the beam and the other three pass 1.00, the same as the open path, so one has of order 6% of attenuating material in the beam and the others have none that the chambers resolve. None passes anything near 0.41, which is what a 2 mm column of buffer behind the same material would give, so none of them holds a liquid

column of the sample thickness. The measurement that passes 0.94 is accordingly the only one of the six that carries scattering from material in the beam path, and that is what the subtraction needs. Fitted by the procedure below, the three that pass 1.00 need a coefficient between 3 and 5 and still leave the 6 °C references 7 to 28% from flat at best, against 3 to 5% for the one that passes 0.94, with the residual peaking near 0.008 Å^{-1}. Averaging the three together does not improve on the best single one, so what they lack is shape and not counting statistics. Whether any of these measurements held buffer changes little over this Q range: the absorption difference is taken up by the coefficient, and the scattering of a 2 mm water column is about 0.0016 mm^{-1}, some 6% of the plateau and flat in Q, so it shifts the level rather than the shape.

How much of that background to subtract is set by the 6 °C state, which is dissolved and whose profile should fall smoothly. Each reference is reduced to the median in each of nine logarithmic Q bins between 0.0036 and 0.0305 Å^{-1}, and two numbers describe the result: the rise, its lowest bin divided by its highest, and the largest fractional increase from one bin to the next going up in Q, which is zero when the profile never turns back up. The coefficient trades one against the other and nothing in these data escapes that trade. At $\alpha = 0.865$, the value that makes the references flattest, the rise is 1.0 but the profile dips 6 to 8% near 0.004 to 0.005 Å^{-1}, and a dip is a subtraction artefact rather than anything the sample does. Removing the dip altogether takes $\alpha = 0.60$, which leaves a two-fold climb toward low Q in its place. We use $\alpha = 0.80$ for all four loadings. The rise is then 1.26 to 1.45 and the profile turns back up by at most 4%, below the scatter of the individual points, and the 6 °C references sit at 0.026 to 0.028 mm^{-1} over 0.006 to 0.03 Å^{-1}.

Figures S7–S10 show $I(Q)$ from the lowest fitted Q bin outward. Below that bin the points sit on the edge of the beamstop shadow, where the background and the sample no longer share a profile. None of this beamtime's background measurements holds a buffer column, as the transmissions above show, so the subtraction cannot take windows, solvent and parasitic scattering off together the way a buffer measured in the same cell does. At 0.004 Å^{-1} the background removed is 2.4 to 3.0 times the 6 °C signal left behind, so subtracting enough to remove the instrumental tail would take the sample's own low-Q signal with it, and the 6 °C references keep a residual rise at every coefficient. A mismatch of the Q scale between the two was tested as the cause and rejected: flattening the residual that way takes a 6% stretch of the background Q axis and improves the agreement by half a percentage point. Averaged over 0.004 to 0.008 Å^{-1}, the 34 fitted groups stand 54 to 322 times the background, so this limit bears on the 6 °C reference alone.

Loading D8 was measured at 6 °C twice, before and after its hold, and Figure S7 shows the later measurement. The earlier one was the first measurement of the beamtime and is not a dissolved state: it carries a $Q^{-2.7}$ excess at low Q that falls to 4% of the plateau above 0.01 Å^{-1}, and it fits $\alpha = 1.15$ against 0.85 to 0.88 for every other reference. The later measurement is the same loading in the same cell, returned to 6 °C after the 28.00 °C hold; it is flat and fits $\alpha = 0.860$. The excess is therefore undissolved material in the earlier measurement rather than a property of the subtraction.

The speckle contrast was measured on a static reference recorded in the same beamtime, as the flat level of its correlation function through the Siegert relation that opens Equation S5. A static sample cannot relax, so each of its correlation functions is flat in delay and sits at $1 + \beta$. The value is *not* taken over the five bins the correlation analysis fits, where that reference reads 0.1226, 0.1323, 0.1355, 0.1372 and 0.1390 with increasing Q: the standard is scratched, and its static parasitic scattering depresses the apparent contrast most strongly at low Q. β is instead read from the plateau the same reference reaches between 0.0085 and 0.0165 Å^{-1}, where nine consecutive bins give 0.1379 to 0.1428 and their inverse-variance mean is $\beta = 0.141$, the single value used for every bin and every group. Using the fitted

bins instead would understate β by 5%, which returns too large an f: the arrested fractions quoted below would rise from 0.93 to 0.96, while the ordering of f in Q and the scaling temperature of Figure 4 would be unchanged.

8.3 Fitting

Equation 2 of the main text was fitted to the five lowest Q bins, 0.00359 to 0.00756 Å^{-1} (175 to 83 nm), with β fixed. A group is fitted when its correlation function is no noisier than a fixed threshold: the median g_2 uncertainty at the lowest Q between 10^{-4} and 10^{-2} s may not exceed 0.016. Two choices follow from what the holds contain. First, p_{fast} is fitted in each group. The holds run from a largely mobile state, where p_{fast} is 0.55 to 1.00, into arrest, where it is 0.20 to 0.51, and a single value shared across a hold is rejected by an F test in every loading, at $p < 10^{-140}$ in the three that arrest. Each fit starts from the minimum of a p_{fast} profile, so that it cannot settle in a solution in which the fast term imitates the slow one. Second, the rate of the fast mode is reported as the fitted coefficient D_0 itself. No mean or median of the stretched decay is formed, because either one is dominated by the tail of the distribution and moves strongly with p_{fast}. Reduced χ^2 is 0.90 to 3.3. Early in arrest τ_{slow} reaches its upper bound of 10^4 s in 17 of 170 bin fits, where the slow mode does not decay within the acquisition; f is set by the height of the plateau and is unaffected.

Fixing p_{fast} would remove one parameter per group, and the data were tested against it. At $p_{\mathrm{fast}} = 1$, where the fast mode is a single exponential, the mean reduced χ^2 of the 28.00 °C hold rises only from 1.18 to 1.41, so a mobile sample does relax exponentially. The three holds that arrest rise from 1.75, 1.89 and 1.64 to 8.27, 9.56 and 7.22, and the fitted curve crosses the data rather than following it. Intermediate fixed values do not recover the fit: averaged over all 34 groups, $p_{\mathrm{fast}} = 1$ gives 6.60, 0.7 gives 3.73 and 0.5 gives 2.44, against 2.21 for the best single value shared within each hold and 1.62 with p_{fast} fitted per group. The broadening of the fast relaxation is therefore a feature of the data in the arresting holds rather than a free parameter absorbing noise.

8.4 Results

Figures S7–S10 show the four loadings in the format of Figure 3 of the main text, with the correlation functions at the four further Q values in panels (d)–(g). Assembly begins before any target is reached. Averaged over $0.004 < Q < 0.008$ Å^{-1}, the scattered intensity rises on the ramp to 2.3 to 3.6 times its level at 20 to 22 °C by 27.5 °C, in every loading.

At 28.00 °C the sample assembles without arresting. The same low-Q intensity grows 22-fold over the hold, to 6.2 mm^{-1}, while f stays at 0.89 or above at every Q for 6500 s.

At the three higher temperatures the sample arrests, and the rate rises steeply with temperature. f at the lowest Q falls to 0.5 after 5810, 1930 and 540 s of hold at 28.25, 28.50 and 28.75 °C. Measured instead from the moment the RTD passed 28.00 °C, which removes the different ramp times, and taking each group at its midpoint, the times are 6240, 2520 and 1280 s, a factor of five over 0.5 °C. In every group in which arrest is under way ($f < 0.85$ at the lowest Q), f increases monotonically with Q: 7 of 7, 7 of 7 and 4 of 4 groups. Figure 4 puts f at the largest length scale for all four holds on one axis. At 28.50 °C the last group gives $f = 0.07$ at 175 nm and 0.16 at 83 nm.

The structure does not set the arrest. At 28.25 °C the first group in which arrest is under way ($t_w = 4140$ s) has an $I(Q)$ between 0.88 and 1.12 times the final 28.00 °C profile at every Q from 0.004

to 0.028 Å^{-1}. At 28.75 °C ($t_w = 6$ s) the ratio is 0.82 to 0.92. That 28.00 °C profile stays mobile. At 28.50 °C arrest sets in at 0.48 to 0.69 of it. Once under way, arrest proceeds while the low-Q intensity keeps rising, by a factor of 1.6 to 4.0 between the first group in which arrest is under way and the last.

The mobile population slows before arrest and then changes far less than the amplitude does. While f at the lowest Q stays at 0.7 or above, D_0 falls 2.7-, 3.8- and 3.4-fold at 28.00, 28.25 and 28.50 °C. Below 0.7 it varies by factors of 1.7, 1.9 and 3.0 at 28.25, 28.50 and 28.75 °C, while $1 - f$ at the lowest Q rises from 0.33, 0.46 and 0.53 to 0.86, 0.93 and 0.84.

9 Code, Data and AI-Assisted Development

9.1 Code and reduced-data availability

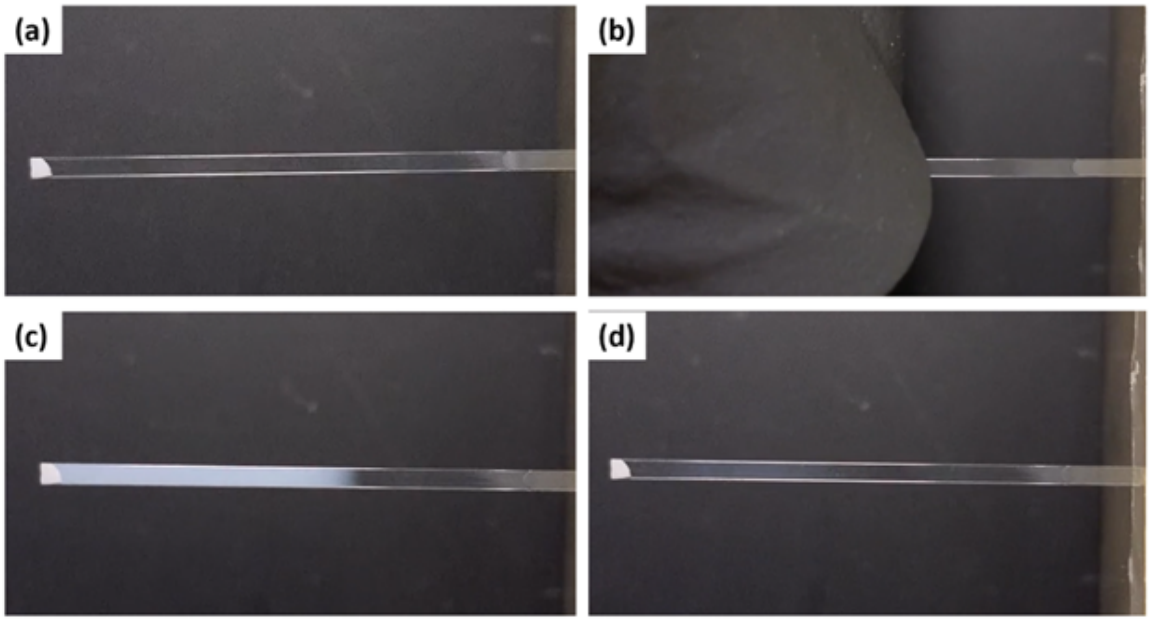
The reduced datasets and all analysis and figure-generation code used for this manuscript are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`. The repository includes the reduced 12-ID-B SAXS/WAXS data, the averaged 8-ID-I XPCS HDF files for every group plotted, the per-acquisition temperature record of the thermal-cycling sequence, radiation-control data, absolute-intensity calibration inputs, and the scripts used to generate the main-text and Supporting Information figures. Every data-derived figure can be regenerated from the repository alone. Two further scripts document how the committed reduced files were produced from the raw beamline results. `SAXS_12id/Read_12ID_SAWAXS.py` handles 12-ID-B, merging the SAXS and WAXS profiles into the committed CSV files. `SAXPCS_8id/average_ranges.py` handles 8-ID-I, applying both outlier-rejection passes to every group on the same terms and averaging each into the committed HDF files. Both read the beamline storage directly. The repository README maps each script and output file to the corresponding manuscript figure. The raw detector frames occupy 3.0 TB and the reprocessed per-acquisition results 51 GB, both held on Advanced Photon Source project storage inside the facility network. The 196 MB of averaged files committed here are what the figures are produced from. Raw data are available from the corresponding authors.

9.2 AI-assisted development

The authors used Claude Sonnet 5 and Opus 5 to assist with the manuscript preparation and code development. The Claude models were accessed through Argo, Argonne National Laboratory's internal generative AI interface, which provides Laboratory staff with access to large language models from multiple vendors, including Anthropic, OpenAI, and Google, for use in Laboratory research and operations.

Qingteng Zhang reviewed the code, reran the analyses, and confirmed the reported results. The code is public (Section 9.1), so every reported number and data-derived figure can be checked independently. All analysis choices, scientific interpretations, and manuscript content were determined by the authors, who take full responsibility for the publication.

Supporting Video



Video S1. Reversible thermal transition in $(VPAVG)_{30}$. (a) A thin quartz capillary containing $(VPAVG)_{30}$ is held in a fixture. (b) The capillary is warmed for approximately 5 s by contact through nitrile gloves. (c) The initially transparent solution becomes visibly turbid. (d) The sample returns to transparency approximately 60 s after removal of the heat source. The complete video can be viewed or downloaded at `https://anl.app.box.com/s/lsukldjnr45nhfrkobijsar1tqhba7jw`.

(a)

A1 (forward)
5′-GTTCCAGCTGTTGGTG

A2 (reverse)
5′-ACCTACTGCTGGTACG

B1 (forward)
5′-TACTTCCAATCCAATGCCGTTCCAGCTGTTGGTGTGCCG

B2 (reverse)
5′-TTATCCACTTCCAATGTTAACCTACTGCTGGTACGCCGAC

$VPAVG_{30}$ OERCA single-stranded template
/5Phos/GTTCCAGCTGTTGGTGTGCCGGCCGTGGGCGTGCCTGC
GGTGGGCGTCCCGGCGGTCGGCGTACCAGCAGTAGGT

(b)

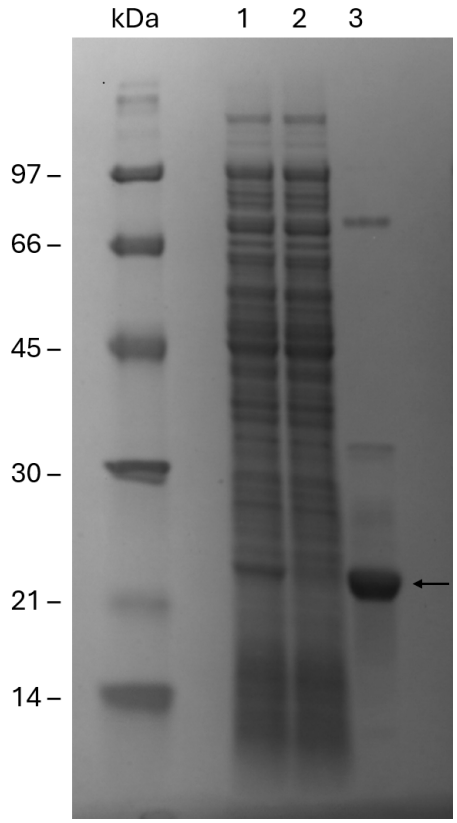


Figure S1: Construction and expression of the $(VPAVG)_{30}$ gene. (a) Oligonucleotide primers and the 5′-phosphorylated single-stranded OERCA template used to construct the ELP gene. (b) SDS-PAGE analysis of expression and purification. Samples were resolved by SDS-PAGE and stained to evaluate expression, solubility, and purity of the heterologously expressed protein. The molecular-weight marker lane is at the left, with prominent bands labeled in kilodaltons (kDa). Lanes 1–3 are the total soluble fraction, the flow-through fraction after IMAC, and the eluted fraction, respectively. The arrow marks the eluted $(VPAVG)_{30}$ construct, whose apparent mobility is lower than that expected from its calculated mass of 15,461 Da.

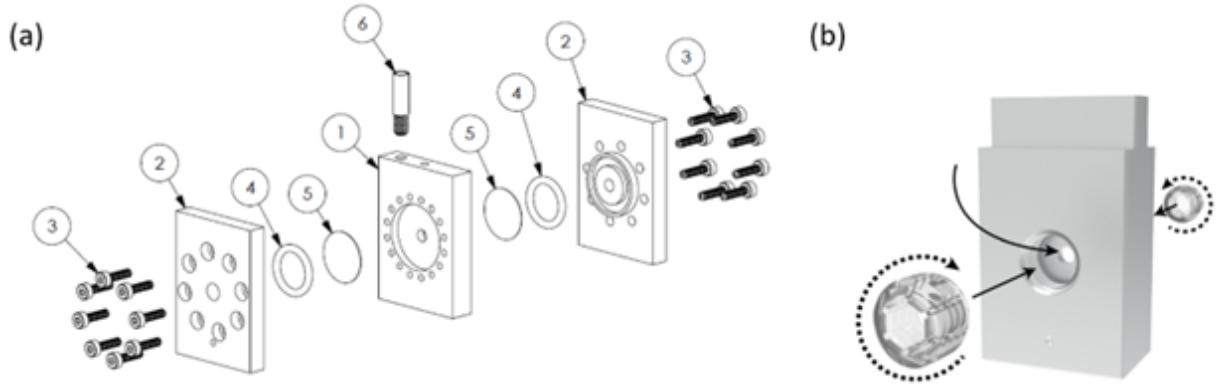


Figure S2: Sample cells used for liquid X-ray measurements. (a) Exploded view of the aluminum cell used for the 8-ID-I SA-XPCS experiments. The cylindrical chamber is 3 mm in diameter and 2 mm thick along the X-ray direction; 0.127 mm polycarbonate windows are compressed by Viton O-rings and bolted caps. During measurement, this cell was inserted into the anodized-aluminum holder of the Peltier-controlled stage described in Section 2.2; both the holder and cell were aluminum. (b) Screw-cap cell developed separately for automated handling; the dashed circles magnify the window seat. Only the bolted aluminum cell in panel (a) was used for measurements reported in the main text. The circled numerals in (a) identify the components: (1) the aluminum cell body, which contains the sample chamber; (2) the aluminum cap plates; (3) M2 screws, 8 mm long, which clamp the caps against the body; (4) Viton O-rings; (5) window disks cut from Kapton or polycarbonate sheet with a hammer-driven punch, polycarbonate being used for the measurements reported here; and (6) the handle used to withdraw the cell from the Quantum Northwest stage. Labels (2)–(5) appear on both sides of the body because the assembly is symmetric about the sample chamber.

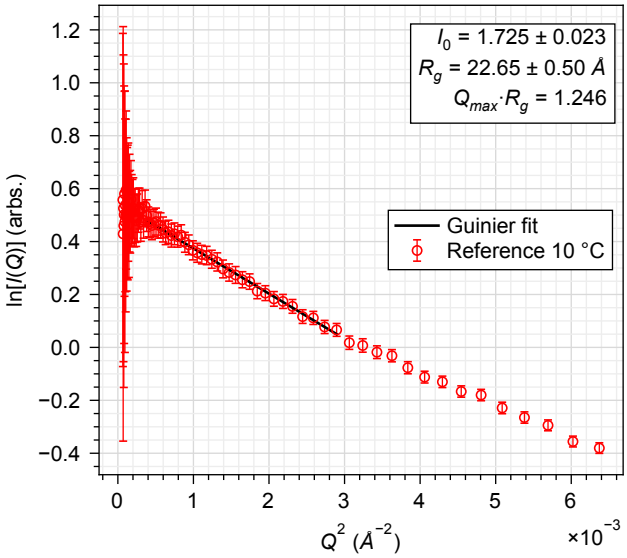


Figure S3: Guinier analysis of the Reference aliquot at 10 °C. Weighted linear fit of $\ln I(Q)$ against Q^2 over $0.020 \le Q \le 0.055$ Å^{-1} (so $Q_{\max} R_g = 1.246$), giving an apparent $R_g = 22.65 \pm 0.50$ Å and $I_0 = 1.725 \pm 0.023$ in the arbitrary intensity units of the reduced 12-ID-B profiles. Points are plotted over the wider range $0.005 \le Q \le 0.080$ Å^{-1} for context; error bars are the propagated intensity uncertainties, σ_I / I.

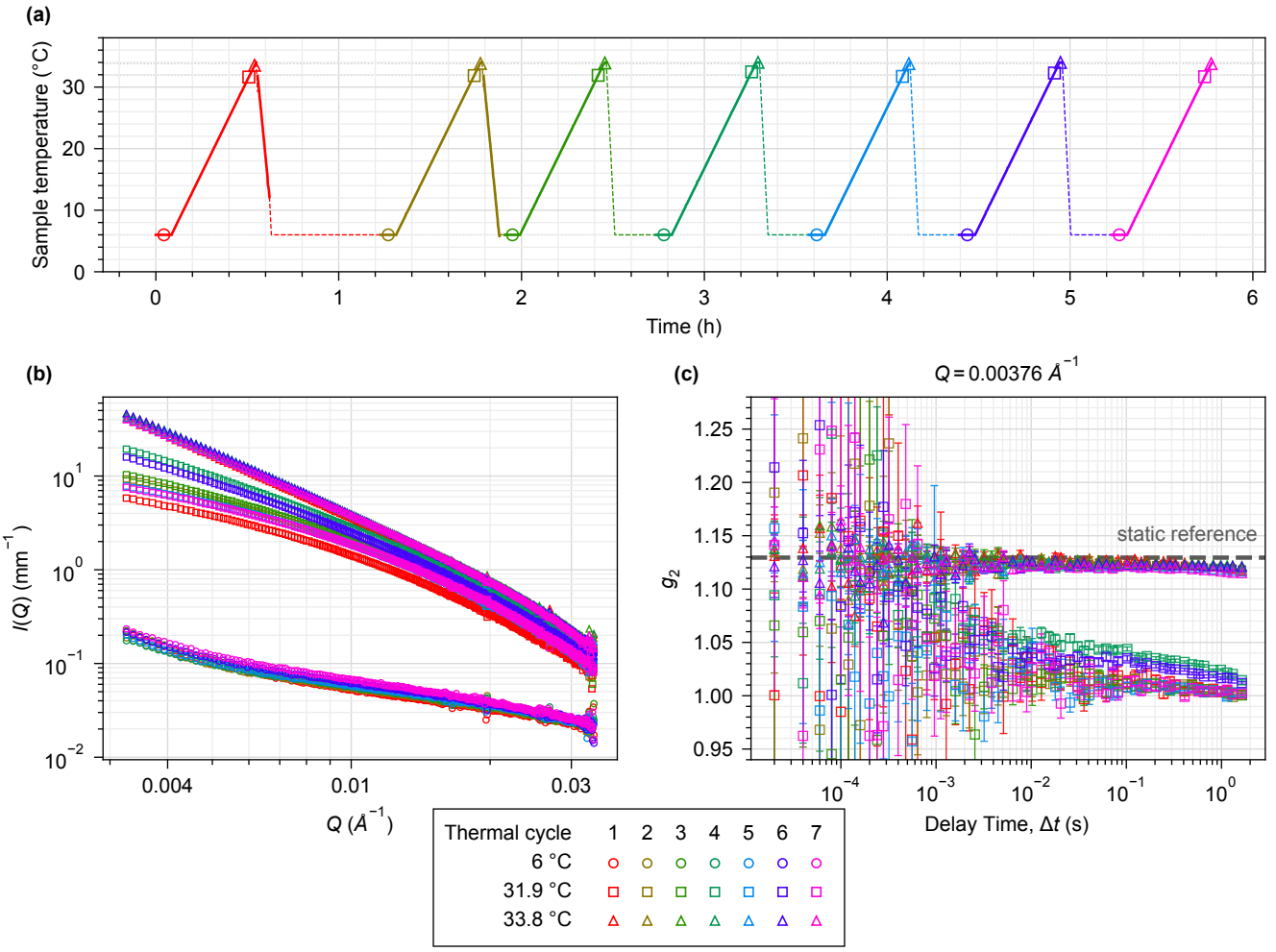


Figure S4: Thermal-cycling repeatability of the SA-XPCS sample: one aliquot through seven consecutive cycles. (a) Sample temperature over the 5.8 h sequence. Solid curves are measured: the stage RTD was read once per 2 s acquisition. No acquisitions were taken while the sample cooled, so no temperature was recorded over those intervals; the dashed curves there are not data but the profile projected from the control protocol, which was fixed at 10 °C min^{-1} cooling until 6 °C followed by a hold at 6 °C until the next ramp. Open symbols mark, at the time and temperature at which it was taken, every acquisition group that is plotted in (b) and (c), so each lower-panel curve can be traced to its point in the thermal history. (b) Buffer-subtracted SAXS on an absolute scale at 6 °C (circles, 32–41 surviving acquisitions averaged; see Section 7.2), 31.9 °C (squares, acquisitions 241–250) and 33.8 °C (triangles, acquisitions 261–270). Each group is placed on the absolute scale with the mean ion-chamber readings of the acquisitions it contains, and the averaged buffer (69 surviving of 89 acquisitions, D0077 and D0080) is subtracted with no empirical scale factor. (c) Correlation functions at the lowest Q bin for the two high-temperature states. The dashed line is the measured correlation level of a static reference recorded in the same beamtime and read at the same Q bin; a static sample cannot relax, so that level is the instrumental contrast. Both are measurements on one axis, so no calibration is imported and none is quoted. The 33.8 °C curves run flat just below the line across four decades of delay, which is arrest; the 31.9 °C curves decay away from it, which is not. Error bars are the measured pointwise g_2 uncertainties; the y axis is capped so that a few short-delay caps clip. One key serves all three panels: its columns are the seven thermal cycles, numbered along the top and colored on the same scale Figure 3 of the main text uses for elapsed time, and its rows are the three temperature states, labeled at the left and distinguished by marker shape.

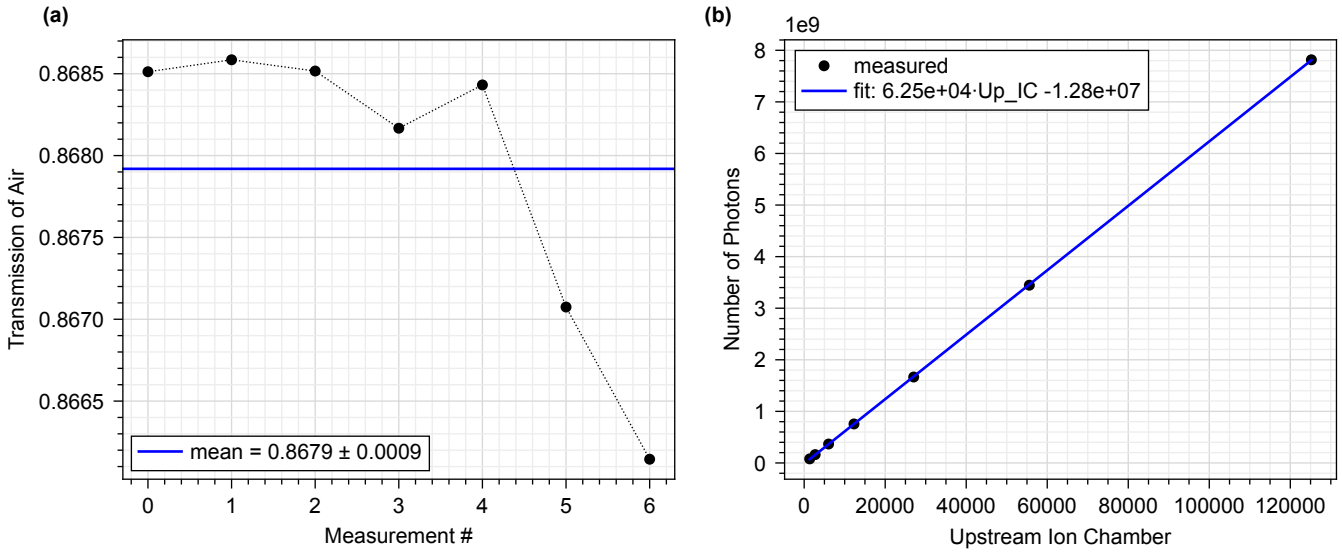


Figure S5: Transmission and photon-flux calibration for conversion to absolute scattering cross section. (a) Air-path transmission across seven measurements in the attenuation series; the line marks the mean, 0.8679 ± 0.0009 (standard deviation). (b) Photon counts recorded by a calibrated PIN diode as a function of the upstream ion-chamber readout (both in the scaled detector units of the calibration series); the line is the linear calibration fit, from which the incident flux follows as $F = (A \cdot \mathrm{Up_IC})/t_{\mathrm{exp}} + B$.

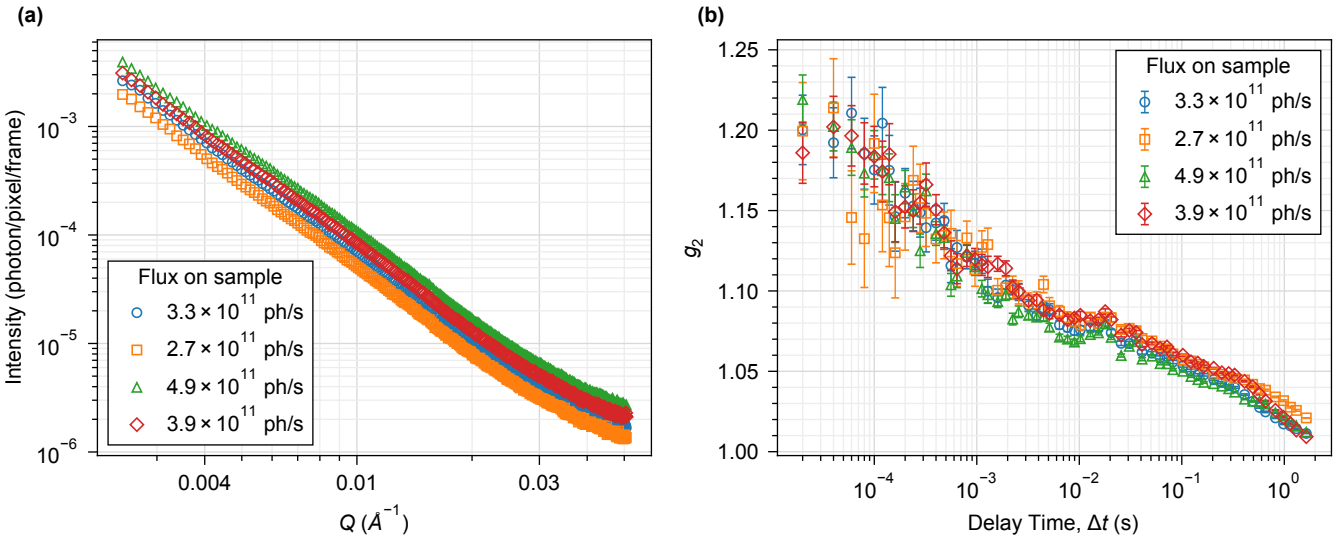


Figure S6: Flux-dependence control for SAXS and SA-XPCS. (a) SAXS profiles and (b) intensity autocorrelation functions from a 20 mg mL^{-1} $(\mathrm{VPAVG})_{30}$ sample at incident fluxes of 2.7×10^{11}–4.9×10^{11} photons s^{-1}. The Q range shown in (a), 0.0024–0.051 Å^{-1}, covers the Q range of Figure 3a of the main text. Intensities in (a) are detector counts per pixel per frame, not on the absolute scale of Figure 3a, so the vertical spacing of the four curves is the incident flux itself: their mean intensities stand in the ratios 1.00, 1.29, 1.58 and 2.01 against incident-monitor ratios of 1.00, 1.24, 1.47 and 1.82. The line shapes nonetheless stay parallel across that range. Panel (b) shows the lowest Q bin of this control configuration, $Q = 0.00304$ Å^{-1} ($2\pi/Q \approx 206$ nm). The legend orders the four conditions by nominal attenuator setting, which is not monotonic in flux. No systematic flux-dependent change is resolved over this range, which substantially exceeds the flux used in the main experiment.

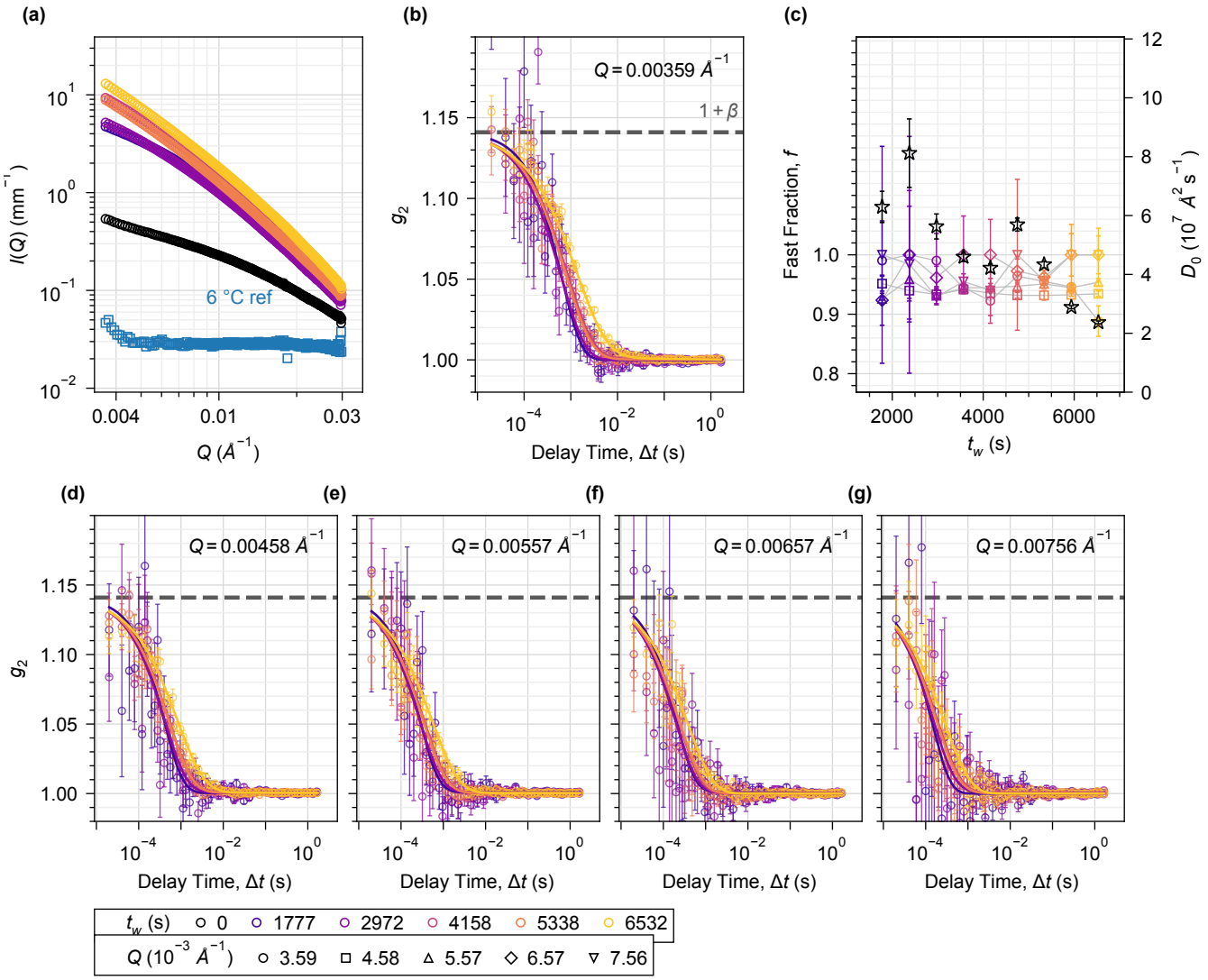


Figure S7: Loading D8 held at 28.00 °C, where it assembles without arresting: the low-Q intensity grows throughout the hold while f stays at 0.89 or above at every Q. (a) Background-subtracted SAXS on the absolute scale of Figure 3a of the main text: the 6 °C profile of loading D8 (blue squares), measured after the hold rather than before it (Section 8.1), the first group at 28.00 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 8.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom; (a), (b) and (d)–(g) show every second group, and the black profile at 0 s is too noisy to fit. Error bars are one-standard-deviation fit uncertainties; Section 8 gives the analysis.

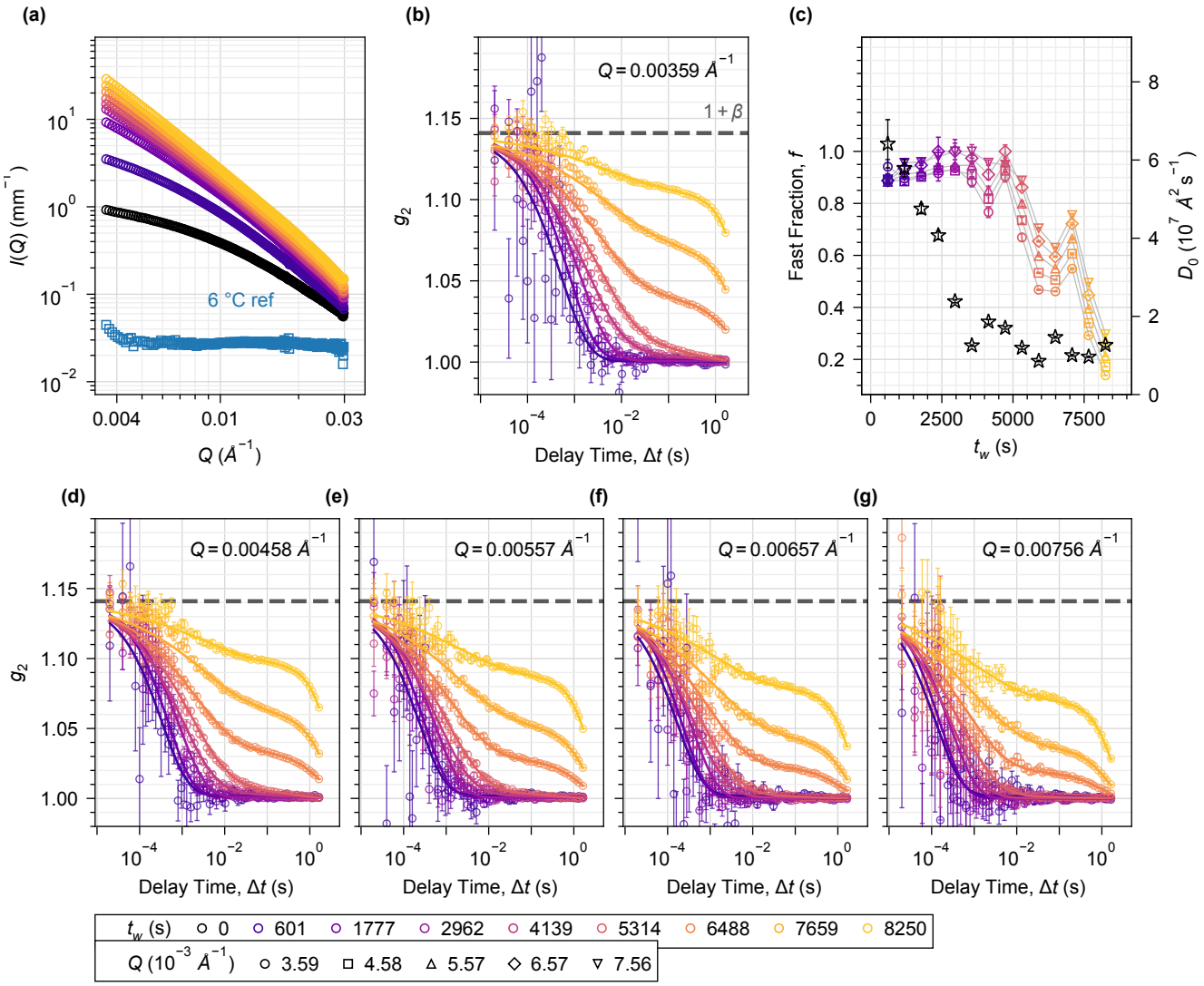


Figure S8: Loading D6 held at 28.25 °C. After about 4000 s the sample arrests, f at the lowest Q falling to 0.14 by 8250 s and increasing with Q at every t_w, while D_0 has already fallen before arrest and then levels off. (a) Background-subtracted SAXS on the absolute scale of Figure 3a of the main text: the 6 °C reference of loading D6 (blue squares), the first group at 28.25 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 8.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom; (a), (b) and (d)–(g) show every second group, and the black profile at 0 s is too noisy to fit. Error bars are one-standard-deviation fit uncertainties; Section 8 gives the analysis.

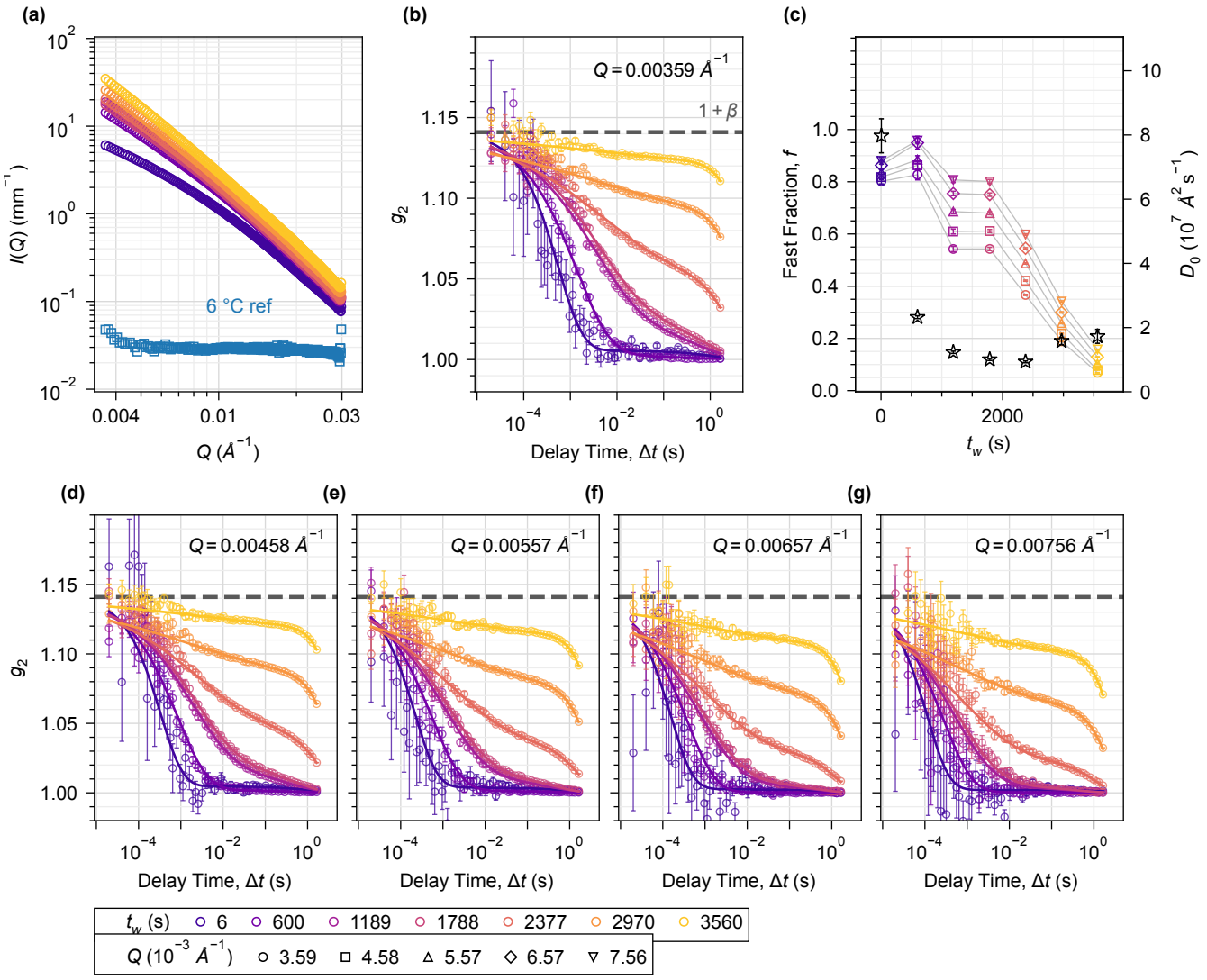


Figure S9: Loading D7 held at 28.50 °C. f at the lowest Q falls from 0.80 to 0.07 within 3600 s, increasing with Q at every t_w. (a) Background-subtracted SAXS on the absolute scale of Figure 3a of the main text: the 6 °C reference of loading D7 (blue squares), the first group at 28.50 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 8.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom. Error bars are one-standard-deviation fit uncertainties; Section 8 gives the analysis.

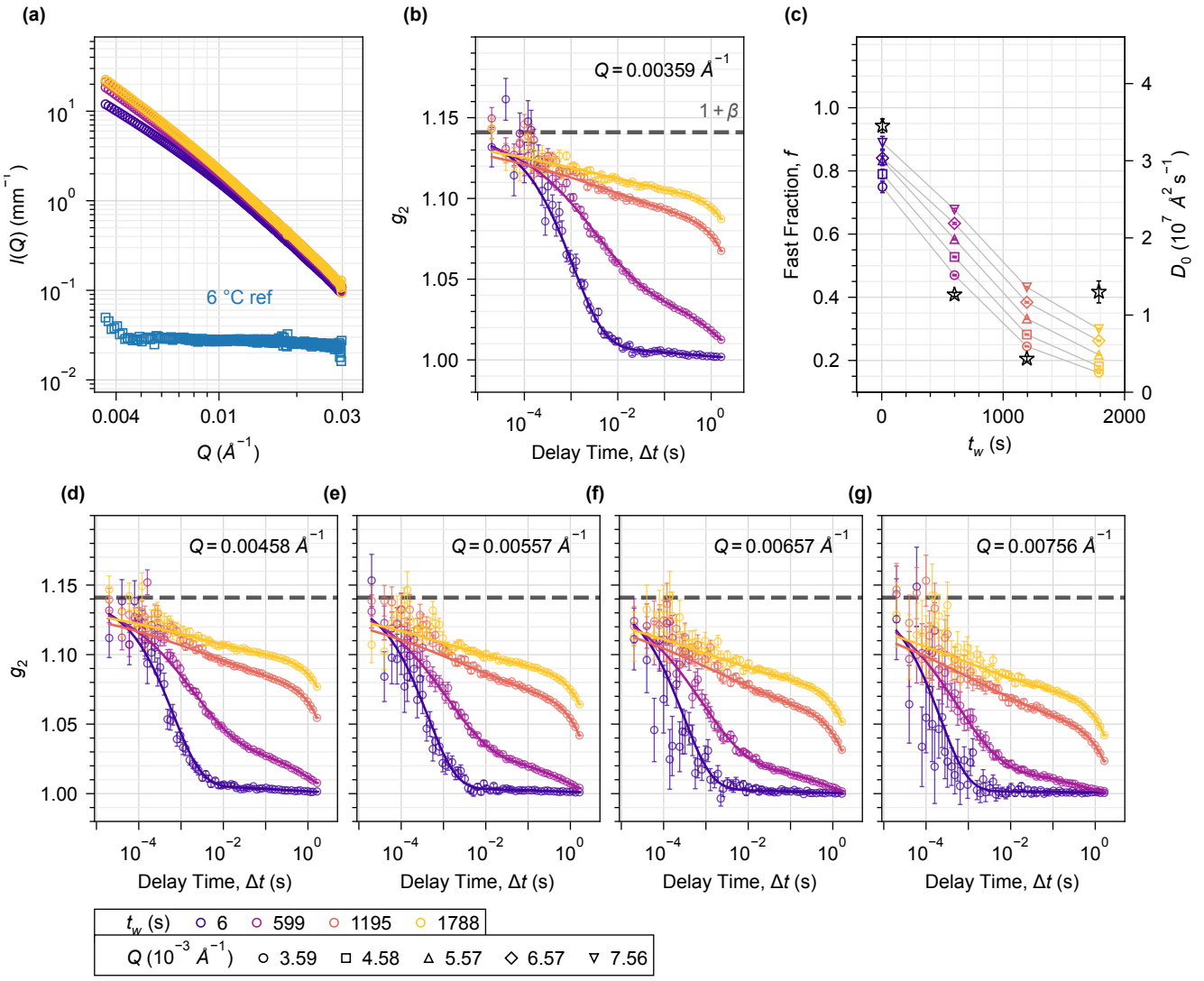


Figure S10: Loading D9 held at 28.75 °C. f at the lowest Q falls from 0.75 to 0.16 within 1800 s, increasing with Q at every t_w. (a) Background-subtracted SAXS on the absolute scale of Figure 3a of the main text: the 6 °C reference of loading D9 (blue squares), the first group at 28.75 °C and later groups. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Equation 2 of the main text at $\beta = 0.141$, p_{fast} free in each group; the dashed line in (b) and (d)–(g) is $1 + \beta$, the contrast of Section 8.2. (c) Left axis, open coloured markers: the fast fraction f at the five Q values, joined by grey guide lines. Right axis, open black stars: the fitted diffusion coefficient of the fast mode, D_0. (d)–(g) g_2 and fits at the four further Q values. Colors denote t_w, keyed along the bottom. Error bars are one-standard-deviation fit uncertainties; Section 8 gives the analysis.

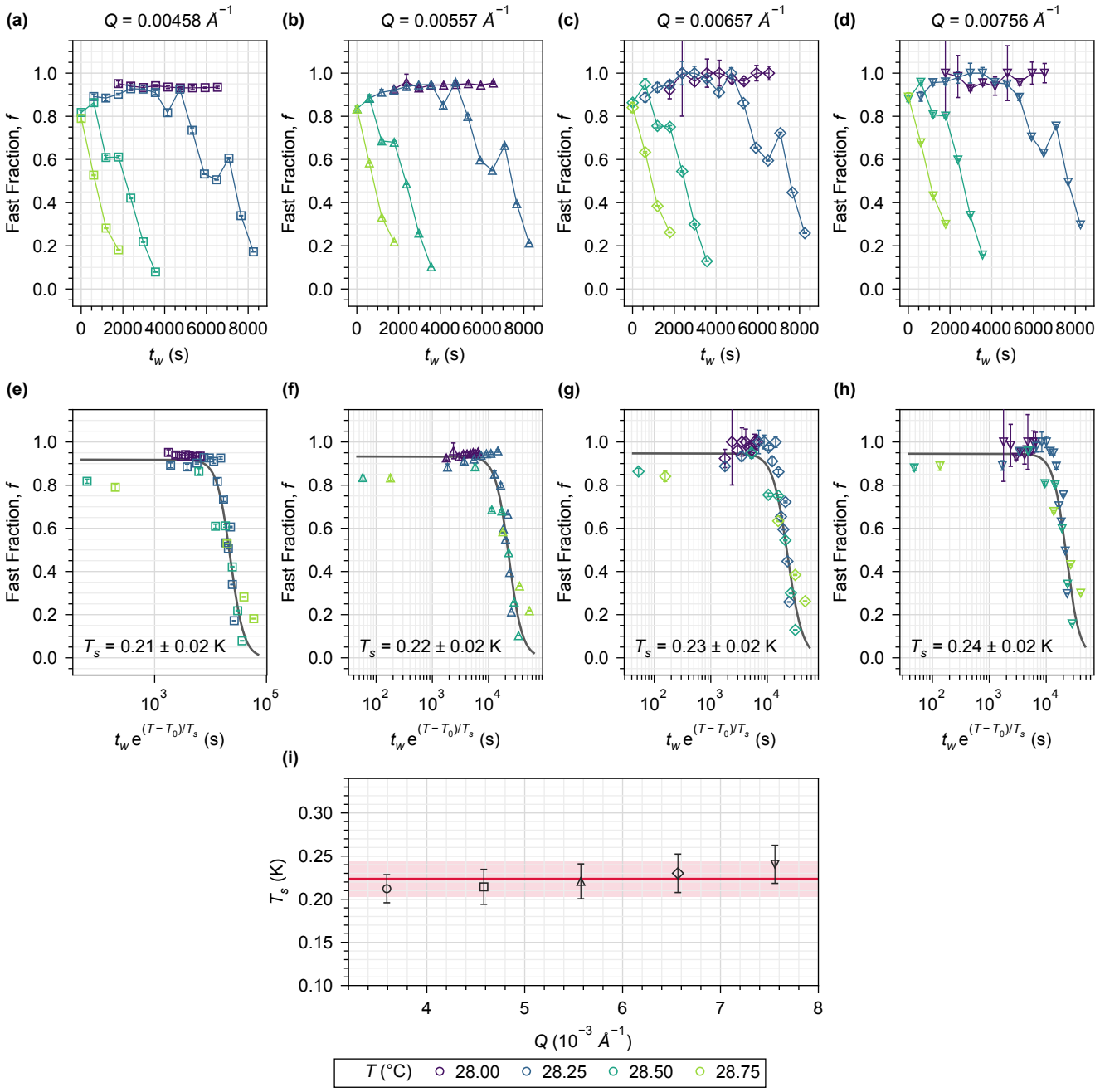


Figure S11: The collapse of Figure 4 repeated at the four higher Q bins. (a)–(d) the fast fraction against t_w; (e)–(h) the same points rescaled by $\exp[(T - T_0)/T_s]$, each panel using the T_s fitted for that bin alone and quoting it with its bootstrap uncertainty, and each with its own master curve. (i) the T_s collapsing each bin alone, against Q, with bootstrap uncertainties; the line is the mean over the five bins and the band the bootstrap uncertainty of a single-bin collapse, $T_s = 0.22 \pm 0.02$ K; the five values themselves spread by only 0.01 K, so the quoted uncertainty is the larger and more conservative of the two. Marker shape denotes Q, colour denotes temperature. The four higher bins return 0.21, 0.22, 0.23 and 0.24 K against 0.21 K at the lowest Q, so one scaling temperature describes every length scale measured to within 13%. Error bars are one-standard-deviation fit uncertainties.

References

[1] Amiram, M.; Quiroz, F. G.; Callahan, D. J.; Chilkoti, A. A highly parallel method for synthesizing DNA repeats enables the discovery of 'smart' protein polymers. *Nat. Mater.* **2011**, *10*, 141–148.

[2] Aslanidis, C.; de Jong, P. J. Ligation-independent cloning of PCR products (LIC-PCR). *Nucleic Acids Res.* **1990**, *18*, 6069–6074.

[3] Eschenfeldt, W. H.; Lucy, S.; Millard, C. S.; Joachimiak, A.; Mark, I. D. A family of LIC vectors for high-throughput cloning and purification of proteins. *Methods Mol. Biol.* **2009**, *498*, 105–115.

[4] Makowska-Grzyska, M.; Kim, Y.; Maltseva, N.; Li, H.; Zhou, M.; Joachimiak, G.; Babnigg, G.; Joachimiak, A. Protein production for structural genomics using E. coli expression. *Methods Mol. Biol.* **2014**, *1140*, 89–105.

[5] Xu, L.; Wang, Z.; Chen, J.; Chen, S.; Yang, W.; Ren, Y.; Zuo, X.; Zeng, J.; Wu, Q.; Sheng, H. Folded Network and Structural Transition in Molten Tin. *Nature Communications* **2022**, *13*, 126.

[6] Zheng, C. Y.; Yao, Y.; Deng, J.; Seifert, S.; Wong, A. M.; Lee, B.; Mirkin, C. A. Confined Growth of DNA-Assembled Superlattice Films. *ACS Nano* **2022**, *16*, 4813–4822.

[7] Ozgulbas, D. Y.; Jensen, D.; Butler, R.; Vescovi, R.; Foster, I. T.; Irvin, M.; Nakaye, Y.; Chu, M.; Dufresne, E. M.; Seifert, S.; Babnigg, G.; Ramanathan, A.; Zhang, Q. Robotic pendant drop: containerless liquid for μs-resolved, AI-executable XPCS. *Light: Science & Applications* **2023**, *12*, 196.

[8] Zhang, Q. et al. 20 µs-resolved high-throughput X-ray photon correlation spectroscopy on a 500k pixel detector enabled by data-management workflow. *Journal of Synchrotron Radiation* **2021**, *28*, 259–265.

[9] Teixeira, J. Small-Angle Scattering by Fractal Systems. *Journal of Applied Crystallography* **1988**, *21*, 781–785.

[10] Sunde, M.; Serpell, L. C.; Bartlam, M.; Fraser, P. E.; Pepys, M. B.; Blake, C. C. F. Common Core Structure of Amyloid Fibrils by Synchrotron X-Ray Diffraction. *Journal of Molecular Biology* **1997**, *273*, 729–739.

[11] Sawaya, M. R.; Sambashivan, S.; Nelson, R.; Ivanova, M. I.; Sievers, S. A.; Apostol, M. I.; Thompson, M. J.; Balbirnie, M.; Wiltzius, J. J. W.; McFarlane, H. T.; Madsen, A. Ø.; Riekel, C.; Eisenberg, D. Atomic Structures of Amyloid Cross-β Spines Reveal Varied Steric Zippers. *Nature* **2007**, *447*, 453–457.

[12] Sheyfer, D.; Zhang, Q.; Lal, J.; Loeffler, T.; Dufresne, E. M.; Sandy, A. R.; Narayanan, S.; Sankaranarayanan, S. K. R. S.; Szczygiel, R.; Maj, P.; Soderholm, L.; Antonio, M. R.; Stephenson, G. B. Nanoscale Critical Phenomena in a Complex Fluid Studied by X-Ray Photon Correlation Spectroscopy. *Phys. Rev. Lett.* **2020**, *125*, 125504.

[13] Switalski, K.; Fan, J.; Li, L.; Chu, M.; Sarnello, E.; Jemian, P.; Li, T.; Wang, Q.; Zhang, Q. Direct measurement of Stokes–Einstein diffusion of Cowpea mosaic virus with 19 μs-resolved XPCS. *Journal of Synchrotron Radiation* **2022**, *29*, 1429–1435.

[14] Bahadur, D.; Zhang, Q.; Dufresne, E. M.; Grybos, P.; Kmon, P.; Leheny, R. L.; Maj, P.; Narayanan, S.; Szczygiel, R.; Swan, J. W.; Sandy, A.; Ramakrishnan, S. Evolution of structure and dynamics of thermo-reversible nanoparticle gels—A combined XPCS and rheology study. *J. Chem. Phys.* **2019**, *151*, 104902.

[15] Chen, Y.; Rogers, S. A.; Narayanan, S.; Harden, J. L.; Leheny, R. L. Microscopic ergodicity breaking governs the emergence and evolution of elasticity in glass-forming nanoclay suspensions. *Phys. Rev. E.* **2020**, *102*, 042619.

[16] Begam, N.; Ragulskaya, A.; Girelli, A.; Rahmann, H.; Chandran, S.; Westermeier, F.; Reiser, M.; Sprung, M.; Zhang, F.; Gutt, C.; Schreiber, F. Kinetics of Network Formation and Heterogeneous Dynamics of an Egg White Gel Revealed by Coherent X-Ray Scattering. *Physical Review Letters* **2021**, *126*, 098001.